

UR5: Jacobian

RBE 501

1 Jacobian

The 6×6 Geometric Jacobian matrix $J(\mathbf{q})$ maps the joint velocities directly to the end-effector spatial velocities. It is constructed column by column based on the current axis orientation vectors $\mathbf{z}_{i-1}$ and link frame origins $\mathbf{o}_{i-1}$ extracted from the forward kinematics steps:

$$T_{01} = e^{[S_1]\theta_1} M_1, \quad T_{02} = T_{01} e^{[S_2]\theta_2} M_2, \quad \dots, \quad T_{06} = T s) \quad (1)$$

Let $\mathbf{o}_{end} = T_{06}(1 : 3, 4)$ represent the instantaneous position of the end-effector tip.

Jacobian Columns Mapping.

$$\text{Column 1 (Joint 1): } J_1 = \begin{bmatrix} \mathbf{z}_0 \times (\mathbf{o}_{end} - \mathbf{o}_0) \\ \mathbf{z}_0 \end{bmatrix}, \quad \text{where } \mathbf{z}_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{o}_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (2)$$

$$\text{Column 2 (Joint 2): } J_2 = \begin{bmatrix} \mathbf{z}_1 \times (\mathbf{o}_{end} - \mathbf{o}_1) \\ \mathbf{z}_1 \end{bmatrix}, \quad \text{where } \mathbf{z}_1 = T_{01}(1 : 3, 3), \quad \mathbf{o}_1 = T_{01}(1 : 3, 4) \quad (3)$$

$$\text{Column 3 (Joint 3): } J_3 = \begin{bmatrix} \mathbf{z}_2 \times (\mathbf{o}_{end} - \mathbf{o}_2) \\ \mathbf{z}_2 \end{bmatrix}, \quad \text{where } \mathbf{z}_2 = T_{02}(1 : 3, 3), \quad \mathbf{o}_2 = T_{02}(1 : 3, 4) \quad (4)$$

$$\text{Column 4 (Joint 4): } J_4 = \begin{bmatrix} \mathbf{z}_3 \times (\mathbf{o}_{end} - \mathbf{o}_3) \\ \mathbf{z}_3 \end{bmatrix}, \quad \text{where } \mathbf{z}_3 = T_{03}(1 : 3, 3), \quad \mathbf{o}_3 = T_{03}(1 : 3, 4) \quad (5)$$

$$\text{Column 5 (Joint 5): } J_5 = \begin{bmatrix} \mathbf{z}_4 \times (\mathbf{o}_{end} - \mathbf{o}_4) \\ \mathbf{z}_4 \end{bmatrix}, \quad \text{where } \mathbf{z}_4 = T_{04}(1 : 3, 3), \quad \mathbf{o}_4 = T_{04}(1 : 3, 4) \quad (6)$$

$$\text{Column 6 (Joint 6): } J_6 = \begin{bmatrix} \mathbf{z}_5 \times (\mathbf{o}_{end} - \mathbf{o}_5) \\ \mathbf{z}_5 \end{bmatrix}, \quad \text{where } \mathbf{z}_5 = T_{05}(1 : 3, 3), \quad \mathbf{o}_5 = T_{05}(1 : 3, 4) \quad (7)$$

The combined configuration formula for the matrix is expressed as:

$$J(\mathbf{q}) = \begin{bmatrix} \mathbf{z}_0 \times (\mathbf{o}_{end} - \mathbf{o}_0) & \dots & \mathbf{z}_5 \times (\mathbf{o}_{end} - \mathbf{o}_5) \\ \mathbf{z}_0 & \dots & \mathbf{z}_5 \end{bmatrix} \quad (8)$$